



Coffee Shop Sales Data Analysis Report

Project Title: Coffee Shop Sales Data Analysis & Business Insights

Tools Used: Microsoft Excel (Pivot Tables, Charts, Data Cleaning, Formulas, Power Query Editor, Excel Date Modals.)

Role: Data Analyst (Project)

1 Project Objective

The objective of this project was to analyse coffee shop transaction data to uncover sales patterns, customer purchasing behaviour, product performance, and peak business hours. The insights derived from this analysis can help improve revenue generation, inventory management, and promotional strategies.

2 Dataset Description

The dataset consists of transactional sales records from a coffee shop.

Key details:

- 3000+ sales transactions of 3 stores
- Data includes daily customer purchases

Columns in dataset:

- Order ID
- Order Number
- Customer Name
- Order Date
- Order Time
- Item Name
- Category (Coffee, Tea, Bakery, etc.)
- Ingredients
- Quantity
- Price
- Total Price

- Payment Mode
 - Store ID
-

3 Data Cleaning & Preparation

To ensure accurate analysis, the following preprocessing steps were performed:

- Removed duplicate records
 - Checked and handled missing values
 - Converted date and time into proper Excel formats
 - Created new calculated columns:
 - **Month** (for monthly trend analysis)
 - **Day of Week**
 - **Hour of Sale**
 - **Total Price = Quantity × Price**
 - Standardized product names for consistency
-

4 Exploratory Data Analysis (EDA)

Excel Pivot Tables and charts were used to explore the data.

📌 Key KPIs Identified

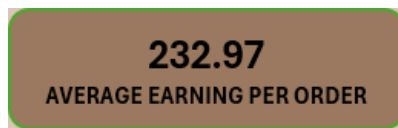
- **Total Revenue**

1817629.00
TOTAL REVENUE

- **Total Orders**

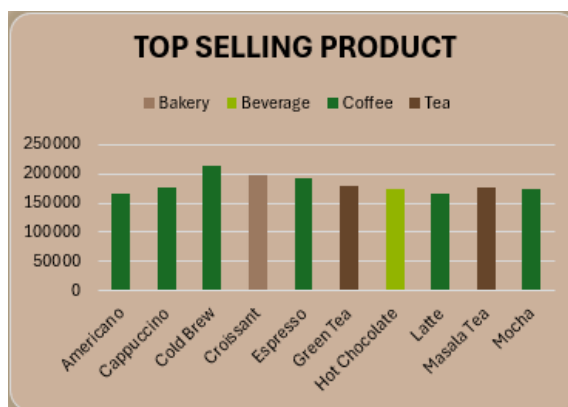
7802
TOTAL ORDERS

- Average Revenue Per Order



Product & Category Analysis

- Identified **best-selling products**



- Identified **low-performing products**
- Compared **revenue by category**

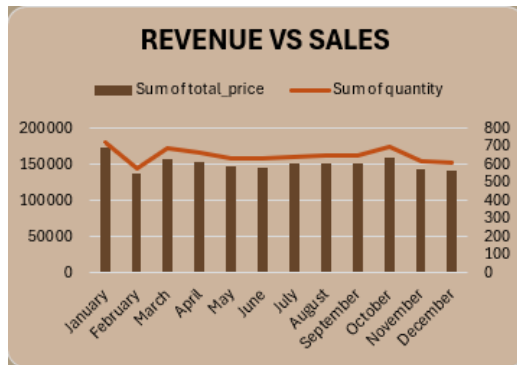


RESULT: Found that the **COFFEE** category contributed most to the sales and in product **COLD BREW** is the most sold item and least sold item is **LATTE and AMERICANO**

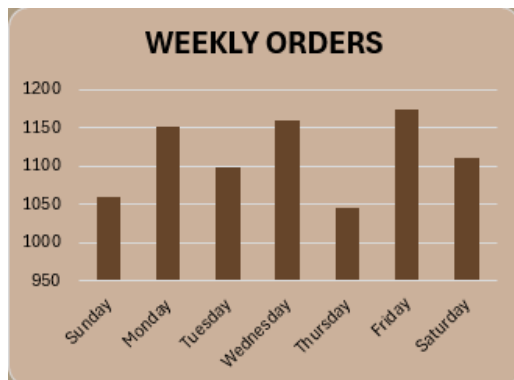


Time-Based Analysis

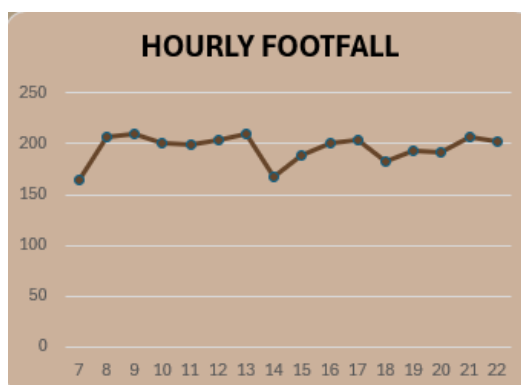
- Monthly sales trends



- Sales comparison by day of the week



- Hourly footfall of the stores



RESULT: Morning and evening time slots showed higher sales activity, **Wednesday** and **Friday** the footfall is maximum, Revenue is decreasing but very slightly

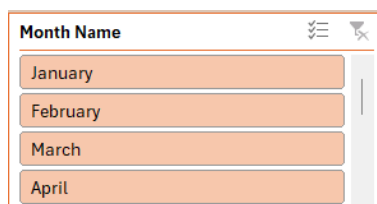
5 Dashboard Creation in Excel

An interactive dashboard was built using:

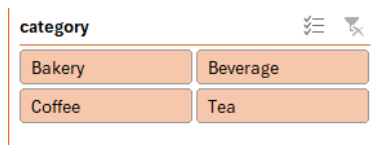
- Pivot Tables

Row Labels	Sum of quantity
Sunday	1060
Monday	1152
Tuesday	1098
Wednesday	1160
Thursday	1046
Friday	1175
Saturday	1111

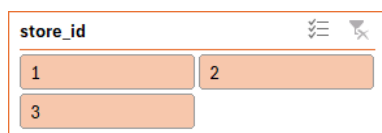
- Pivot Charts
- Slicers for filtering by:
 - Month



- Category



- Stores

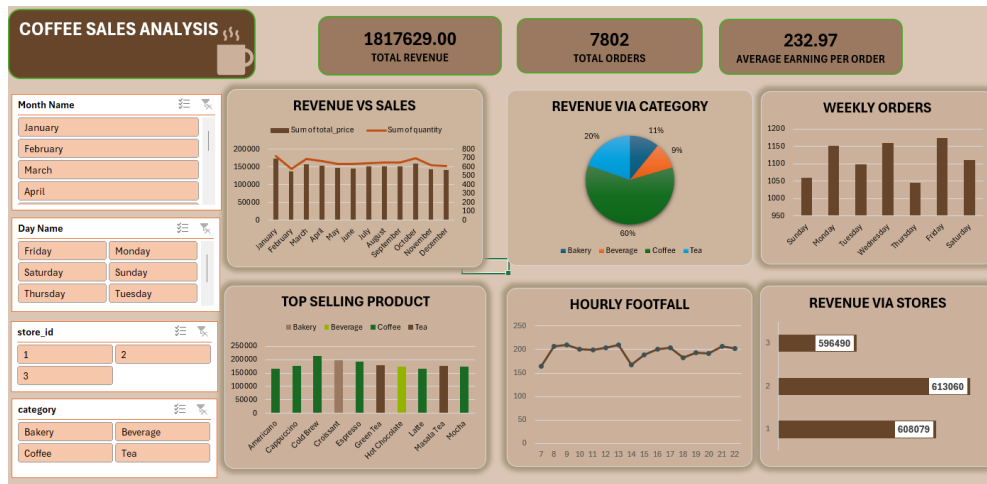


Charts included:

- Line chart → Hourly Footfall
- Bar chart → Yearly Revenue and Orders, Top Sold Product with Category , Weekly Orders, Revenue via different Stores.
- Pie chart → Revenue via Category

- Column chart → Revenue via Stores

6 Key Business Insights



1. Coffee category contributes the highest share of total revenue.
2. Sales peak during morning hours and in evening especially after 4pm indicating strong demand for breakfast beverages.
3. Weekdays show higher average order value specially on Monday, Wednesday and Friday as compared to weekends.
4. A few products have consistently low sales and may require promotion.
5. Store 2 contributes maximum to the sales as compared to other 2

7 Business Recommendations

- Increase stock and staff during peak morning hours as well as after 4pm
- Introduce combo offers eg : with Latte we can offer Croissant
- Run promotional campaigns during low-sales hours
- Focus marketing efforts on Coffee
- Some event initiative should be taken on weekends some painting classes or pottery to increase the footfall on weekends (talk to the people who organises events .)
- For retention the coffee shop should work on its ambience so that the people will get attracted and the remote workers would prefer to sit there.

Skills Demonstrated

Technical Skills (Excel):

- Data Cleaning
- Pivot Tables
- Data Visualization
- Dashboard Creation
- Formulas & Functions (SUMIF, COUNTIF, etc.)
- Power Query Editor
- Dax Formulas

Analytical Skills:

- Sales Trend Analysis
- KPI Identification
- Business Insight Generation
- Data-driven Decision Support

Conclusion

This project demonstrates

All the 3 stores are doing great, to increase the revenue we should offer combos, focus on peak hours by increasing staff as well as inventory, more focus on weekdays because office going people or students are the consumers
